

Centralized Audit Trail Compliance Engine

SE Lab 2 - Agile Backlog Creation & Sprint Simulation in Jira

SUBMISSION SNAPSHOT

Project	Centralized Audit Trail Compliance Engine
Jira space / board	Centralized Audit Trail Compliance Engine - Lab 2 / CATCEL2 board
Actors	Compliance Officer; Security Auditor
Epics	4
User Stories	12
Planned Story Points	60
Sprint 1	5 stories / 26 points / completed
Sprint 2	7 stories / 34 points / completed

Purpose

This document consolidates the Jira Agile backlog, sprint execution evidence, Burndown Chart, and reflection required for Lab 2. The work continues the same Centralized Audit Trail Compliance Engine defined in Lab 1.

Evidence note

Actual Jira screenshots are included to demonstrate the completed backlog, Sprint 1 and Sprint 2 boards/reports, and the Burndown Chart.

1. Project Overview & Lab Objectives

PROJECT CONTEXT

The Centralized Audit Trail Compliance Engine is a compliance logging platform that ingests enterprise application event logs, masks sensitive PII in accordance with GDPR, and generates cryptographically verifiable immutable audit exports.

Lab 2 converts the functional requirements from Lab 1 into Agile backlog items, prioritizes and estimates them, then simulates two Scrum sprints and analyses progress.

Lab requirements covered

- Jira backlog with Epics and User Stories.
- Story point assignments using Fibonacci-style estimates.
- Sprint 1 and Sprint 2 execution through To Do -> In Progress -> Done.
- Burndown Chart showing remaining work against the guideline.
- Reflection on estimation, prioritization, sprint alignment, and capacity.

Sprint plan

Sprint	Goal	Stories	Points
Sprint 1	Build the secure foundation for audit logging.	5	26
Sprint 2	Enable audit investigation and verifiable reporting.	7	34

2. Epics

Epic	Name	Description
EPIC-1	Audit Log Ingestion & Protection	Enable secure ingestion, validation, PII masking, and integrity-protected storage of enterprise audit logs.
EPIC-2	Audit Search & Investigation	Enable authorized users to search, filter, and investigate historical audit records efficiently.
EPIC-3	Immutable Audit Export	Enable users to create cryptographically verifiable and immutable audit exports for compliance and security investigations.
EPIC-4	Security & Access Control	Provide role-based access control and protection of sensitive audit information.

Epic totals

Epic	Stories	Points
Audit Log Ingestion & Protection	4	21
Audit Search & Investigation	3	11
Immutable Audit Export	3	18
Security & Access Control	2	10
TOTAL	12	60

Estimation approach

The backlog uses Fibonacci-style story points of 3, 5 and 8 to represent relative effort, complexity and uncertainty.

3. User Story Register

Key	Epic	Story / User Story	Priority	Pts	Sprint
CATCEL2-5	Audit Log Ingestion & Protection	Ingest Audit Logs As a Compliance Officer, I want the system to ingest structured JSON audit logs, so that enterprise events can be captured for compliance monitoring.	High	5	Sprint 1
CATCEL2-6	Audit Log Ingestion & Protection	Validate Audit Logs As a Compliance Officer, I want audit logs to be validated before processing, so that malformed or incomplete records do not enter the audit system.	High	3	Sprint 1
CATCEL2-7	Audit Log Ingestion & Protection	Mask Sensitive PII As a Compliance Officer, I want sensitive PII such as Social Security Numbers and credit card numbers to be masked, so that personal information is protected.	High	5	Sprint 1
CATCEL2-8	Audit Log Ingestion & Protection	Store Audit Record with Integrity Hash As a Security Auditor, I want each audit record to be stored with a SHA-256 integrity hash, so that I can verify that the record has not been altered.	High	8	Sprint 1
CATCEL2-9	Audit Search & Investigation	Search Audit Records As a Security Auditor, I want to search historical audit records, so that I can investigate security and compliance events.	High	3	Sprint 2
CATCEL2-10	Audit Search & Investigation	Filter Audit Records As a Security Auditor, I want to filter audit records by date, application, event type, severity, and actor, so that I can quickly find relevant events.	High	5	Sprint 2

Jira keys, parent Epics, priorities, point values and sprint allocation are reproduced from the completed board evidence.

3. User Story Register (continued)

Key	Epic	Story / User Story	Priority	Pts	Sprint
CATCEL2-11	Audit Search & Investigation	View Integrity Status As a Security Auditor, I want to see the integrity status of audit records, so that I can identify potentially modified records.	Medium	3	Sprint 2
CATCEL2-12	Immutable Audit Export	Generate Audit Export As a Compliance Officer, I want to generate an audit export containing selected records, so that I can provide evidence during compliance reviews.	High	5	Sprint 2
CATCEL2-13	Immutable Audit Export	Verify Record Integrity Before Export As a Security Auditor, I want the system to verify record integrity before exporting, so that corrupted or modified records are detected.	High	5	Sprint 2
CATCEL2-14	Immutable Audit Export	Seal Immutable Export As a Compliance Officer, I want the generated audit export to be immutable, so that the evidence cannot be silently modified.	High	8	Sprint 2
CATCEL2-15	Security & Access Control	Role-Based Access As a Compliance Officer or Security Auditor, I want role-based access control, so that only authorized users can access audit functions.	High	5	Sprint 1
CATCEL2-16	Security & Access Control	Protect Sensitive Audit Data As a Compliance Officer, I want sensitive audit information to remain protected, so that unauthorized users cannot access PII.	High	5	Sprint 2

Jira keys, parent Epics, priorities, point values and sprint allocation are reproduced from the completed board evidence.

4. Jira Backlog Evidence - Part 1

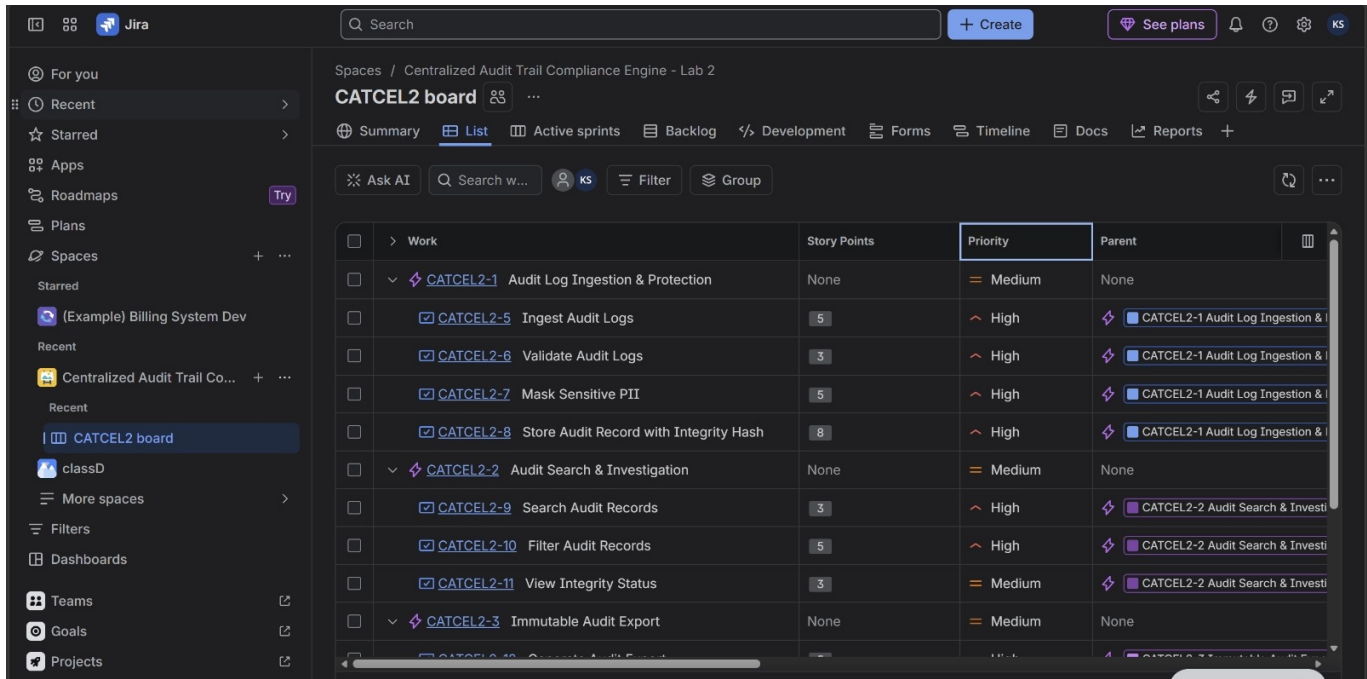


Figure 1. Jira List view showing the first portion of the completed backlog with Epic grouping, story points, priority, and parent relationships.

4. Jira Backlog Evidence - Part 2

The screenshot shows the Jira interface for the 'CATCEL2 board' within the 'Centralized Audit Trail Compliance Engine - Lab 2' space. The left sidebar contains navigation options like 'For you', 'Recent', 'Starred', 'Apps', 'Roadmaps', 'Plans', 'Spaces', 'Filters', 'Dashboards', 'Teams', 'Goals', and 'Projects'. The main area displays a list of backlog items with columns for checkboxes, work items, story points, priorities, and parent items. The items are grouped by epic, with some items expanded to show sub-tasks.

	Work	Story Points	Priority	Parent
☐	CATCEL2-9 Search Audit Records	3	High	CATCEL2-2 Audit Search & Investi
☐	CATCEL2-10 Filter Audit Records	5	High	CATCEL2-2 Audit Search & Investi
☐	CATCEL2-11 View Integrity Status	3	Medium	CATCEL2-2 Audit Search & Investi
☐	CATCEL2-3 Immutable Audit Export	None	Medium	None
☐	CATCEL2-12 Generate Audit Export	5	High	CATCEL2-3 Immutable Audit Expo
☐	CATCEL2-13 Verify Record Integrity Before Export	5	High	CATCEL2-3 Immutable Audit Expo
☐	CATCEL2-14 Seal Immutable Export	8	High	CATCEL2-3 Immutable Audit Expo
☐	CATCEL2-4 Security & Access Control	None	Medium	None
☐	CATCEL2-15 Role-Based Access	5	High	CATCEL2-4 Security & Access Co
☐	CATCEL2-16 Protect Sensitive Audit Data	5	High	CATCEL2-4 Security & Access Co

Figure 2. Jira List view showing the remaining backlog items and Epic groupings with story points and priorities.

5. Sprint 1 - Active Sprint Evidence

Goal: Build the secure foundation for audit logging. **Scope:** 5 stories / 26 points.

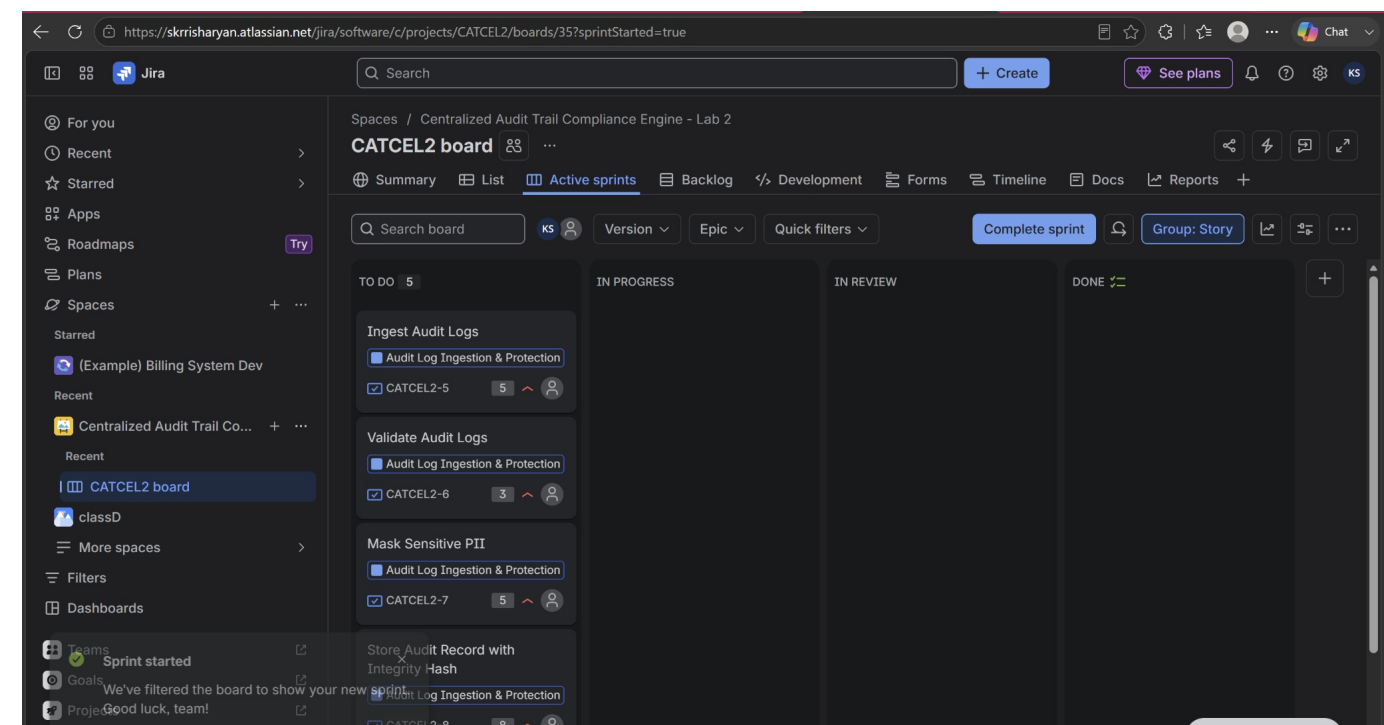


Figure 3. Jira Active Sprint view for Sprint 1, showing the five planned stories on the Scrum board.

5. Sprint 1 - Completion Evidence

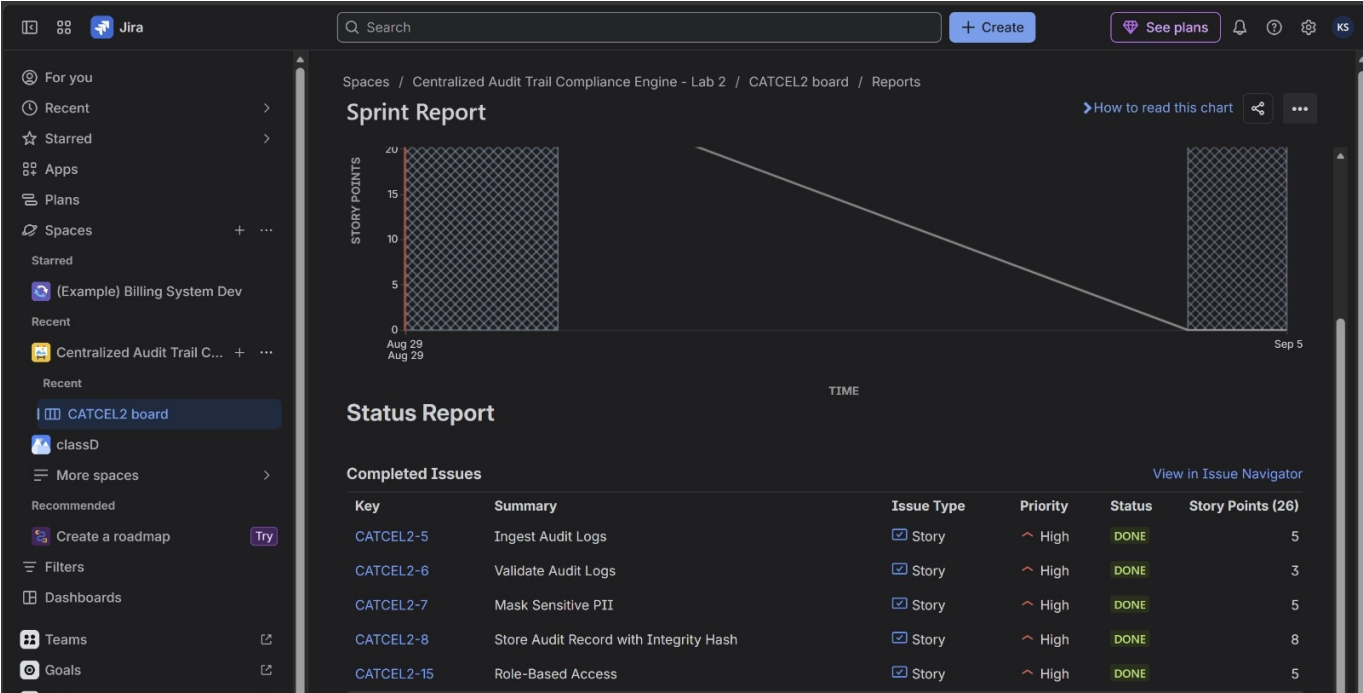


Figure 4. Jira Sprint Report for Sprint 1, showing the completed issues and 26 story points.

6. Sprint 2 - Active Sprint Evidence

Goal: Enable audit investigation and verifiable reporting. **Scope:** 7 stories / 34 points.

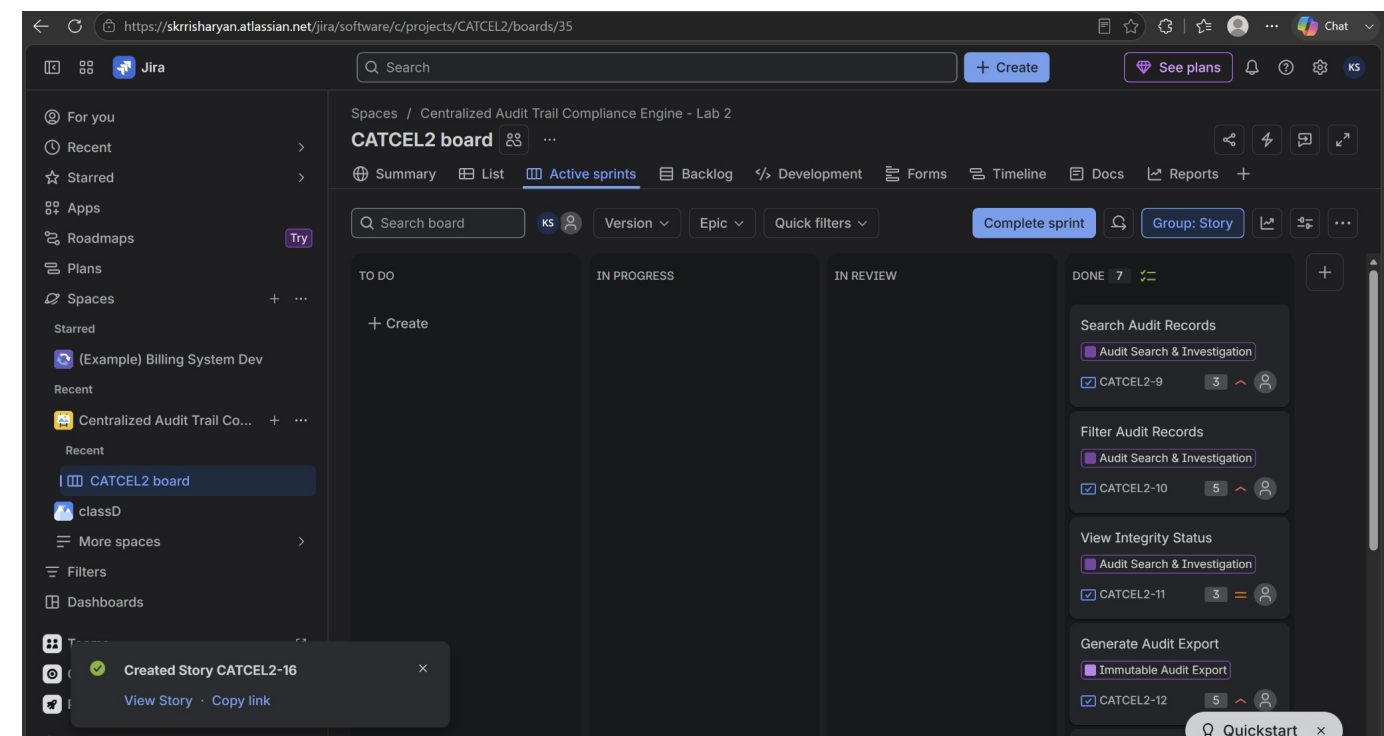


Figure 5. Jira Active Sprint view for Sprint 2, showing all seven stories in the Done column.

6. Sprint 2 - Completion Evidence

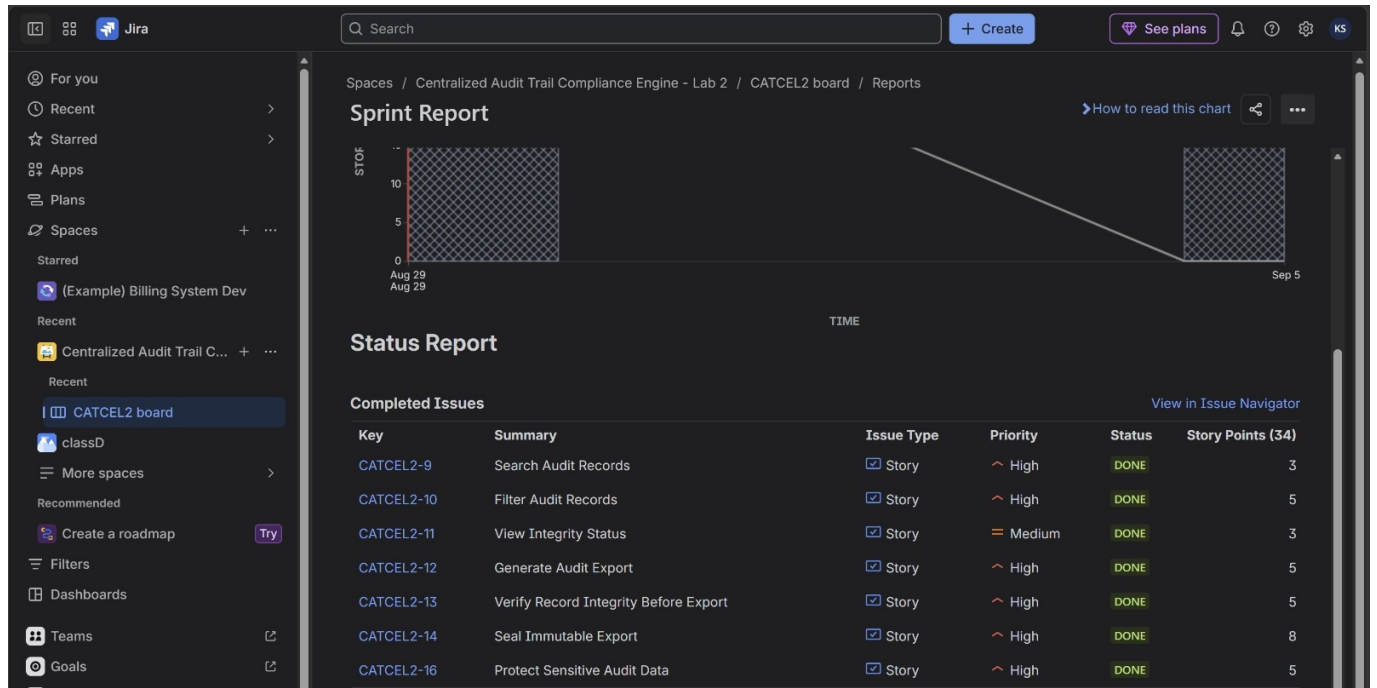


Figure 6. Jira Sprint Report for Sprint 2, showing the completed issues and 34 story points.

7. Burndown Chart & Analysis

The following is the actual Jira Burndown Chart for Sprint 2, using Story Points as the estimation statistic. It starts at 34 points and reaches 0 by the end of the simulated sprint.



8. Reflection & Final Checklist

1. Did your estimations reflect the actual effort?

Yes. The estimates were reasonably aligned with the simulated effort. Smaller stories such as searching and validating audit logs were estimated at 3 points, while more complex work such as integrity hashing and sealing immutable exports were estimated at 8 points. The Fibonacci scale helped represent increasing complexity and uncertainty.

2. Was your backlog well-prioritized?

Yes. Foundational features such as audit-log ingestion, validation, PII masking, integrity protection, and access control were prioritized before investigation and export features. This created a logical implementation order for the compliance engine.

3. How did your simulated sprint align with your plan?

The simulated sprints followed the planned structure. Sprint 1 contained five foundation stories totaling 26 points, while Sprint 2 contained seven investigation, export, and security stories totaling 34 points. All planned stories were progressed through To Do -> In Progress -> Done.

4. What insights did the burndown chart give about your team's capacity?

The Sprint 2 burndown started with 34 story points and reached 0 remaining points by the end of the sprint. This indicates that the planned Sprint 2 workload was completed within the simulation and suggests that a workload of approximately 34 story points was achievable in this exercise.

FINAL SUBMISSION CHECKLIST

- ✓ 4 Epics created and documented.
- ✓ 12 User Stories created and linked to the correct Epics.
- ✓ Priorities and Fibonacci story points assigned.
- ✓ Sprint 1 completed with 26 points.
- ✓ Sprint 2 completed with 34 points.
- ✓ Active Sprint evidence captured for both sprints.
- ✓ Burndown Chart captured from Jira.
- ✓ Reflection questions answered.

Appendix - Lab 1 to Lab 2 Traceability

Lab 1 requirement	Lab 2 mapping
FR-001 Ingest structured JSON audit logs	Epic 1 - Ingest Audit Logs; Validate Audit Logs
FR-002 Mask sensitive PII	Epic 1 - Mask Sensitive PII
FR-003 Persist with SHA-256 integrity hash	Epic 1 - Store Audit Record with Integrity Hash
FR-004 Search and filter historical audit records	Epic 2 - Search Audit Records; Filter Audit Records; View Integrity Status
FR-005 Generate immutable verifiable exports	Epic 3 - Generate Audit Export; Verify Record Integrity; Seal Immutable Export
NFR-002 Security & privacy	Epic 4 - Role-Based Access; Protect Sensitive Audit Data